TT-M-261D
March 11, 1970
SUPERSEDING
Int. Fed. Spec. TT-M-00261C
December 30, 1964 and
Fed. Spec. TT-M-261B
August 26, 1959

FEDERAL SPECIFICATION

METHYL ETHYL KETONE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Methyl Ethyl Ketone covered by this specification is the 99-percent grade for use primarily as a solvent or thinner in protective coatings.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-P-143 - Paints, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
PPP-D-729 - Drum: Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-D-732 - Drums, Metal, 55-Gallon Reconditioned (For Shipment of Noncorrosive Material).
PPP-P-704 - Pails, Metal: (Shipping, Steel 1 Through 12 Gallon).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 791/9601 - Lubricants, Liquid Fuels, and Related Projects; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specification, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distributions in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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Laws and Regulations:

Code of Federal Regulations:

Title 49, Parts 71 - 78, inclusive - Transportation.
21 CFR Part 191 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 270 - Petroleum and Petroleum Products, Sampling.
D 740 - Standard Specifications for Methyl Ethyl Ketone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Quantitative and qualitative requirements. The Methyl Ethyl Ketone shall conform to the requirements of ASTM D 740.

3.2 Labeling. Labeling shall conform to the Federal Hazardous Substances Act.

3.3 Workmanship. The Methyl Ethyl Ketone shall be visually free of sediment or suspended matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container; or manufactured in a plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

4.2.2 Package lot. An indefinite number of 55-gallon drums or smaller unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of material from one isolated container; or filled with a homogeneous mixture of material manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

4.3 Sampling.

4.3.1 Sampling for tests. Take sample for tests in accordance with ASTM D 270. Test the sample in accordance with ASTM D 740.

4.3.1.1 Packaged lot. Take a random sample of packaged containers from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) of 2.5 percent defective. Examine the sample in accordance with 4.4.1.

4.4 Inspection. Perform inspection in accordance with method 9601 of Fed. Test Method Std. No. 791.

4.4.1 Examination of the preparation for delivery. Examine samples taken in accordance with 4.3.1 for compliance with MIL-STD-290, Fed. Std. No. 123 or TT-P-143 (see 5.1, 5.2, and 5.3), with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or

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underfilled containers exceeds the acceptance number required in MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests. All tests cited in ASTM D-740 are quality conformance tests.

5. PREPARATION FOR DELIVERY

5.1 Civil agencies.

5.1.1 Packaging and packing. The Methyl Ethyl Ketone shall be packaged in 1-gallon cans, 5-gallon pails or 55-gallon drums. Level shall be A, B, or C, as specified (see 6.2).

5.1.1.1 One gallon cans. The one gallon quantities shall be packaged and packed in cans conforming to TT-P-143.

5.1.1.2 Five-gallon pail or 55-gallon drum. The 5-gallon or 55-gallon quantities shall be packaged in pails conforming to PPP-P-704, type I, class 3, 4, or 5 or in drums conforming to PPP-D-729 or PPP-D-732, type I or II, respectively.

5.1.1.3 Bulk. When specified (see 6.3), Methyl Ethyl Ketone shall be furnished in bulk quantity in accordance with the Department of Transportation Regulations.

5.2 Military agencies.

5.2.1 Packaging, packing and marking shall be in accordance with MIL-STD-290.

5.3 Marking. In addition to markings required by the contract or order, the bottles or cans and shipping containers shall be marked in accordance with Fed. Std. No. 123 and DOT regulations.

5.3.1 Special marking. The following statement with a caution label shall appear prominently on each container, and shall conform to the requirements of the Fed. Hazardous Substances Act:

METHYL ETHYL KETONE

WARNING! FLAMMABLE

Keep away from heat and open flame.
Keep container closed.
Use with adequate ventilation.
Avoid prolonged breathing of vapor.
Avoid prolonged or repeated contact with skin.

6. NOTES

6.1 Intended use and properties of the material. Methyl Ethyl Ketone ($CH_3COC_2H_5$) covered by this specification is intended for use in surface coatings and in paint and varnish removers. Compared to acetone it has essentially the same solvent power, is less soluble in water, has a higher boiling point and a slower evaporation rate. Main intended uses are as low boiling active solvent in nitrocellulose lacquers, vinyl lacquers, paint and varnish removers, and as a solvent for cellulose acetate and cellulose acetate butyrate in aircraft dopes. It has good solvency for most natural and synthetic resins and is useful for the formulation of high solids lacquers. It has the characteristic ketone odor but leaves no residual odor. The flash point is about 24° F.

6.2 Methyl Ethyl Ketone may be purchased either (a) by volume, the unit being a gallon of 231 cubic inches at 20° C. (68° F.), or (b) by weight. A gallon of Methyl Ethyl Ketone at 20° C. (68°F.), weighs approximately 6.71 pounds. The exact weight in pounds per gallon of any sample can be determined by multiplying the specific gravity at 20°/20° C. (68°/68° F.), by 8.33. If the specific gravity at 20° C. is 0.806, the weight per gallon at this temperature will be 0.806 x 8.33 = 6.714 pounds.

6.3 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity required.
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Special marking required (see 5.3.1).

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MILITARY CUSTODIANS:

Army - MU
Navy - WP
Air Force - 68

Review activities:

Army - MU, MO, MR
Navy - WP, SH
Air Force - 68

User activities:

Army - GL, MO
Navy - GG, YD

Preparing activity:

GSA-FSS

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